

```
import math
```

```
def calculate_azithromycin(total_dose_per_day, total_days, times_per_day,
double_first_day=False):
```

```
    """
```

```
    Calculate azithromycin dosage for different formulations
```

```
    """
```

```
    print("\n=== Azithromycin Dosage Calculations ===")
```

```
    # Adjust for double first day if needed
```

```
    if double_first_day:
```

```
        print("NOTE: Patient needs double dose on first day")
```

```
        total_medication = total_dose_per_day * (total_days + 1)
```

```
    else:
```

```
        total_medication = total_dose_per_day * total_days
```

```
    # Calculate and print formulations
```

```
    print_formulations(total_medication, total_days, times_per_day)
```

```
def print_formulations(total_medication, total_days, times_per_day):
```

```
    """Print dosage options for different azithromycin formulations"""
```

```
    # 100mg/5ml suspension
```

```
    calc_suspension(total_medication, total_days, times_per_day, strength=100,
                    bottle_sizes=[15, 30])
```

```
    # 200mg/5ml suspension
```

```
    calc_suspension(total_medication, total_days, times_per_day, strength=200,
                    bottle_sizes=[15, 30])
```

```
    # 250mg tablets
```

```
    calc_tablets(total_medication, total_days, times_per_day, strength=250)
```

```
    # 500mg tablets
```

```
    calc_tablets(total_medication, total_days, times_per_day, strength=500)
```

```
def calc_suspension(total_med, total_days, times_per_day, strength, bottle_sizes):
```

```
    """Calculate suspension formulation dosage"""
```

```
    dose_per_time = math.floor((total_med * 5 / strength) / (times_per_day * total_days))
```

```
    bottles = [math.ceil((total_med * 5 / strength) / size) for size in bottle_sizes]
```

```
    print(f"\nAzithromycin {strength}mg/5ml Suspension:")
```

```
    print(f"- Dose: {dose_per_time}ml every {24//times_per_day} hours")
```

```
    print(f"- Duration: {total_days} days")
```

```
    print(f"- Bottles needed: {bottles[0]} ({bottle_sizes[0]}ml) OR {bottles[1]} ({bottle_sizes[1]}
ml)")
```

```

def calc_tablets(total_med, total_days, times_per_day, strength):
    """Calculate tablet formulation dosage"""
    tabs_per_day = math.ceil(total_med / (strength * total_days))
    total_tabs = math.ceil(total_med / strength)

    print(f"\nAzithromycin {strength}mg Tablets:")
    print(f"- Dose: {tabs_per_day} tablet(s) every {24//times_per_day} hours")
    print(f"- Duration: {total_days} days")
    print(f"- Total tablets needed: {total_tabs}")

def mycoplasma_pneumonia_treatment():
    """
    Main function for Mycoplasma pneumonia treatment decision support
    """
    print("\n=== Mycoplasma Pneumonia Treatment Calculator ===")
    print("Please enter patient details:\n")

    try:
        # Get patient information
        weight = float(input("Patient weight (kg): "))
        age = int(input("Patient age (years): "))

        # Get treatment frequency (default once daily)
        times_per_day = int(input(
            "\nEnter frequency (1-3 doses/day), or press enter for default (1): ") or 1)

        # Calculate total daily dose (5mg/kg/day, max 250mg for adults)
        dose_per_day = min(5 * weight, 250) if age >= 12 else 10 * weight

        # Get treatment duration (default 5 days)
        total_days = int(input(
            "\nEnter treatment duration (3-7 days), or press enter for default (5): ") or 5)

        # Print treatment recommendation
        print("\n=== Treatment Recommendation ===")
        print(f"Azithromycin {dose_per_day}mg daily for {total_days} days")

        # Calculate and display formulations
        calculate_azithromycin(dose_per_day, total_days, times_per_day,
                               double_first_day=(age >= 12))

        # Print important notes
        print("\n=== Important Notes ===")
        print("- Take medication at the same time each day")
    
```

```
if age >= 12:
```

```
    print("- Take double dose on first day (500mg for adults)")
```

```
print("- Complete full course even if symptoms improve")
```

```
print("- May cause gastrointestinal side effects")
```

```
# Disclaimer
```

```
print("\n=== DISCLAIMER ===")
```

```
print("This tool is for educational purposes only.")
```

```
print("It does not replace clinical judgment.")
```

```
print("Always consult a physician for medical decisions.")
```

```
except ValueError:
```

```
    print("\nError: Please enter valid numeric values")
```

```
if __name__ == "__main__":
```

```
    mycoplasma_pneumonia_treatment()
```